SOC Lab Report: Brute Force Attack Detection using Splunk

Lokeshwar V

1 Abstract

This report demonstrates the simulation and detection of a brute force attack in a SOC environment using Splunk. The attack was performed using Kali Linux against a Windows 11 system, and logs were analyzed to detect malicious behavior. The process includes attack simulation, detection, investigation, response, and reporting aligned with MITRE ATT&CK.

2 Objective

To simulate a brute force attack and detect it using Splunk SIEM, followed by investigation and response actions.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Initial Access
- Technique: Brute Force (T1110)

5 Attack Simulation

The brute force attack was simulated using Hydra from Kali Linux targeting the RDP service.

5.1 Command Used

```
hydra -l administrator -P rockyou.txt rdp://192.168.56.50
```

5.2 Explanation

This command attempts multiple password combinations against the Administrator account using the RDP protocol.

5.3 Screenshot

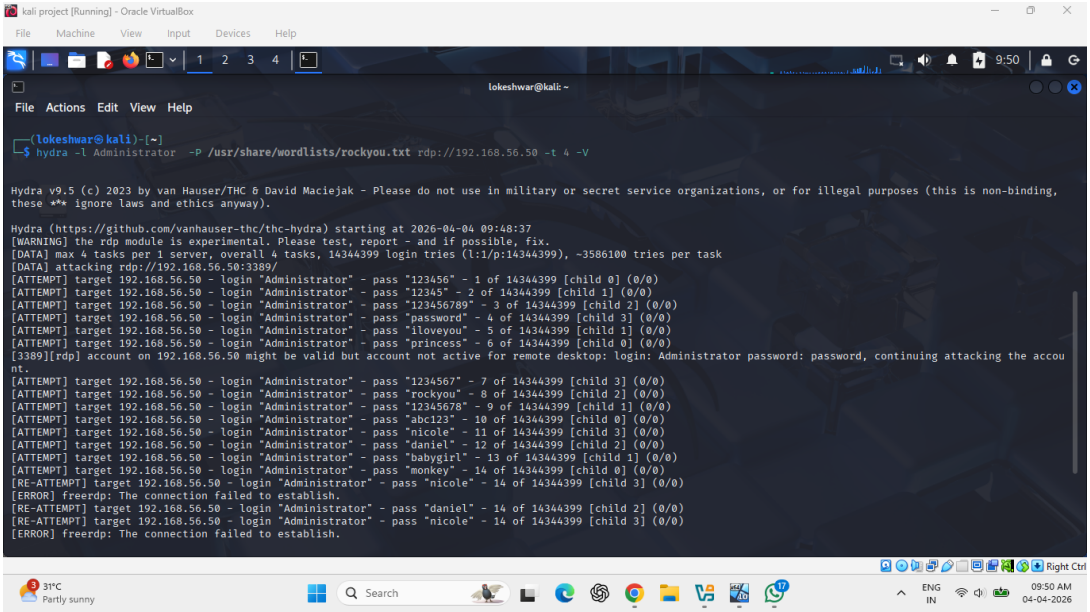

Figure 1: Brute Force Attack using Hydra

6 Detection in Splunk

Failed login attempts are recorded using EventCode 4625 in Windows Security logs.

6.1 Detection Query

```
index=windows EventCode=4625
| bin _time span=5m
| stats count by Source_Network_Address, _time
| where count > 10
| sort -count
```

6.2 Explanation

The query identifies IP addresses generating excessive failed login attempts within a short time frame.

6.3 Screenshot

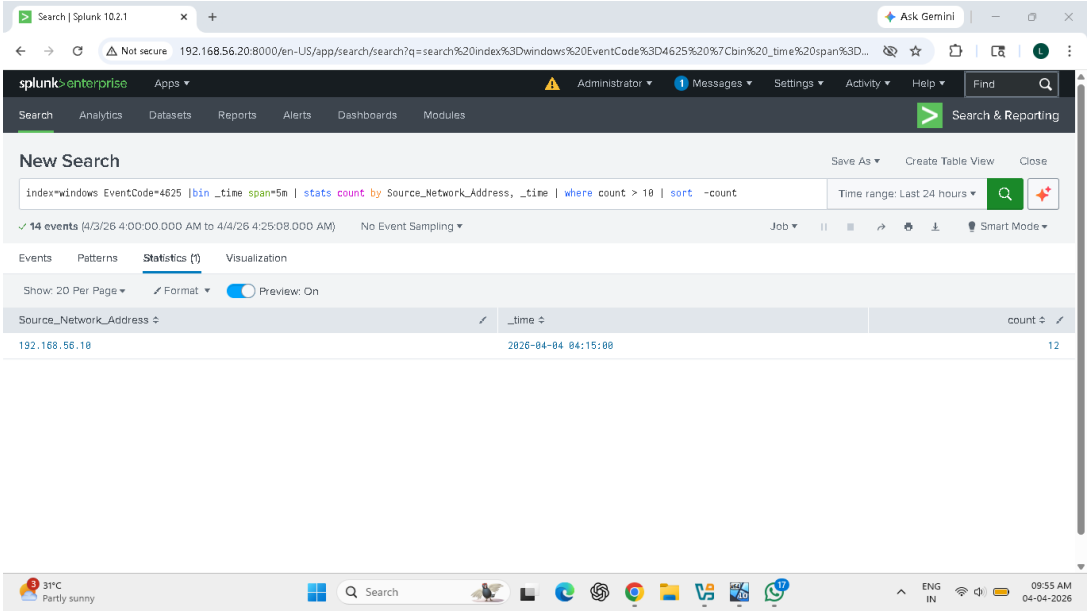

Figure 2: Detection of Brute Force Attack in Splunk

7 Investigation

Further analysis was conducted to identify attacker details and targeted accounts.

7.1 Investigation Query

```
index=windows EventCode=4625
| stats count by Account_Name, Source_Network_Address
| sort -count
```

7.2 Findings

- Attacker IP: 192.168.56.X
- Target Account: Administrator
- High number of failed login attempts

7.3 Screenshot

8 Response Actions

Immediate mitigation steps were taken.

- Blocked attacker IP using firewall

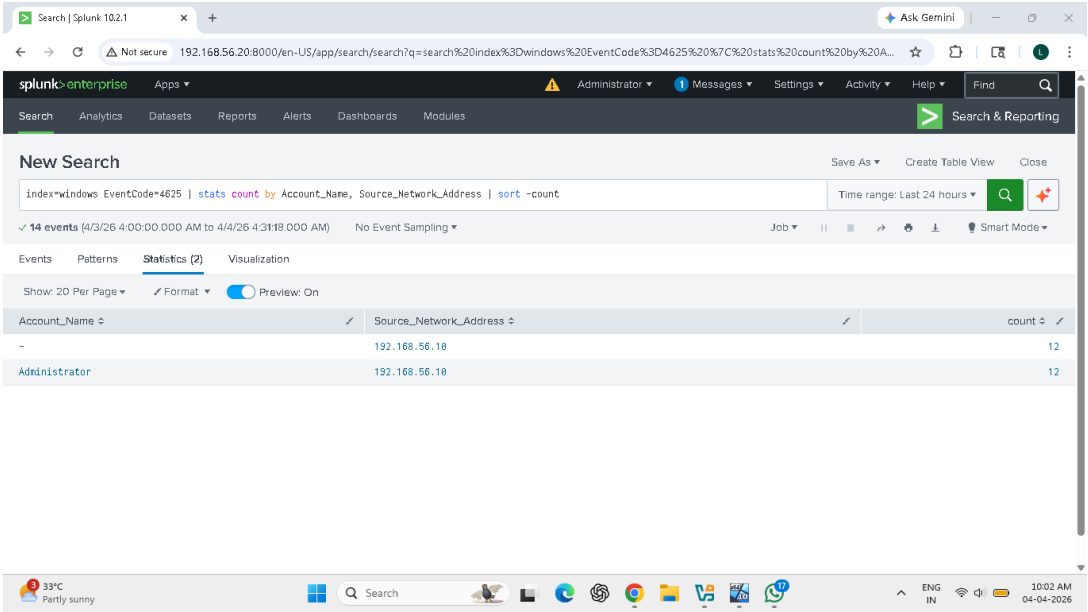

Figure 3: Investigation Results showing attacker IP and target user

- Disabled the targeted account temporarily
- Enabled continuous monitoring

8.1 Commands Used

```
New-NetFirewallRule -DisplayName "Block-Attacker" -Direction Inbound -Remo
net user administrator /active:no
```

8.2 Explanation

The attacker IP was blocked to prevent further access attempts. The Administrator account was temporarily disabled to secure the system.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

A brute force attack was detected targeting the Administrator account via RDP using multiple failed login attempts.

9.2 Impact

The attack posed a risk of unauthorized access to the system.

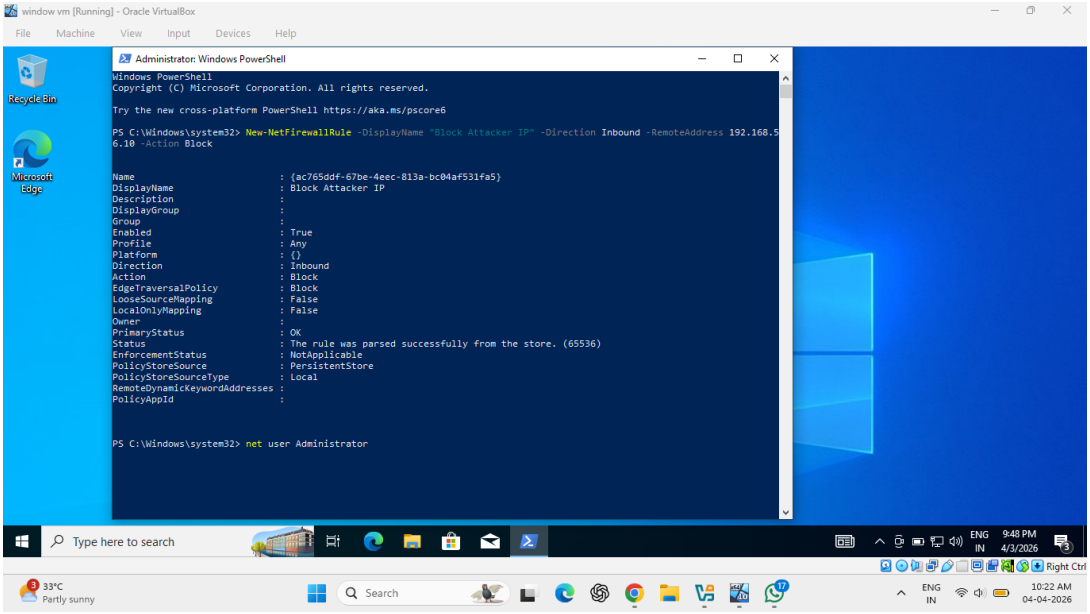

Figure 4: Response Actions - Blocking IP and Disabling Account

9.3 Conclusion

The attack was successfully detected using Splunk and mitigated through appropriate response actions. This demonstrates the effectiveness of SIEM in SOC operations.

10 Conclusion

This experiment highlights how brute force attacks can be detected and handled efficiently using Splunk. It also emphasizes the importance of timely detection, investigation, and response in a SOC environment.